

GDEV 32

Final Project Specifications

The final project is a culmination of everything that you have learned in this class. You will expand your semester-long demo program (that you started working on since Exercise 1) to implement additional graphical features.

The following are possibilities for features you can implement (along with the grade you can expect for implementing them correctly and completely):

B-tier (max grade of 85) – at least two of the following:

- Transparency (via alpha blending)
- Reflections via environment maps
- Soft shadow rendering (hint: this part is basically an advanced version of the upcoming Exercise 3)

B+-tier (max grade of 90) – must meet B-tier first, then add one of the following:

- Reflections via mirror planes
- Refractions (not just reflections) via environment maps
- Stylistic lighting – cel-shading, other non-photorealistic rendering, or whole-scene fog
- Post-processing framebuffer effects – film grain, motion blur, depth of field, chromatic aberration, or bloom

A-tier (max grade of 95) – must meet B+-tier first, then add 2 additional B+-tier effects (total of 3).

* each category above only counts once (e.g., bloom + film grain will not count as two separate effects)

* you cannot get more than 95 for implementing more than 3 B+-tier effects

* if you implement only 1 additional effect, your max grade will be between 90-95

S-tier (max grade of 100) – must meet A-tier first, then add one of the following:

- Advanced realistic lighting — self-shadowing normal maps, parallax mapping, ambient occlusion, etc.
- Advanced stylistic lighting – subsurface scattering, volumetric fog, “Breath of the Wild”-style deferred rendering, etc.
- Real-time reflections or refractions via SSR (screen-space reflection) techniques
- Decal rendering – bullet holes, spray paint, etc. (decals must be dynamically placed by the application user, e.g., firing a gun); must perform proper wall clipping/wrapping for full credit
- Procedural generation of graphical assets, e.g., foliage, landscapes, or cityscapes (subject to the instructor’s approval; must be significantly complex!)
- Physics simulation (subject to the instructor’s approval; must be significantly complex!)
- Any other sufficiently advanced OpenGL-based effect (subject to the instructor’s approval)

Note that the specified grades are “maximum possible” – you may not necessarily get this maximum score if your feature’s implementation is incomplete or buggy.

Submission Requirements

1. You need to submit a **proposal** for your final project on Canvas. Make sure to list your proposed features in bullet-point form. You may use mockup diagrams to help explain your vision for the project. The deadline for your proposal will be announced in class.
 - If you specify only features that lead to a B or B+ project, you will forfeit the possibility of getting a higher grade.
 - You won't get credit if you implement an effect that you did not originally propose.
 - Any change in your proposed features after the proposal deadline is subject to approval by the instructor, and may incur deductions in your final grade (if the replacement features are less complex than originally proposed).
 - Tip: When writing your proposal, try to specify what you plan to do in order of difficulty. Then, when developing your project, aim for the features in order -- from least difficult to most difficult. This strategy doubles as insurance in case you run out of time to produce an A-tier/S-tier project.
2. The final defense date will be announced later on Canvas, and you will be required to **sign up for a defense slot**. During the defense, I will ask detailed questions about your implementation of your graphical effects.
3. Kindly prepare a **short video** (2 minutes maximum) of your application running, in case you can't get your project to work during the defense. (You can skip this if you're confident, but do so at your own risk.)
4. A **.zip file** of your project and video should be uploaded on Canvas.
5. As usual, you are **not allowed to use any external C++ libraries** other than those already provided to you in gdevgl.zip. You may, however, use external programs to help generate your assets (e.g., Blender, Python scripts, etc.), which may then be loaded by your final project's C++ program. Any external source should be mentioned in your Certificate of Authorship.
6. A **final peer evaluation** should be submitted after your defense. You will rate your partner on a scale of 0.00 (worst) to 1.00 (best), and you will need to provide an explanation for your rating. This peer evaluation will be posted on Canvas as an assignment.